

# Optimization – 3

The Euler–Lagrange equation and the direct method

Homework

10 сентября 2026 года

**Theoretical minimum.** The Euler–Lagrange equation and its two first integrals; natural boundary and transversality conditions; the multiplier for an isoperimetric constraint; convexity in  $(y, y')$  as the sufficient condition; completing the square for quadratic functionals; the direct method — minimising sequence, compactness, lower semicontinuity — and the ways it fails.

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1. Prove that a solution of the Euler–Lagrange equation is a global minimiser when  $F$  is convex in  $(y, y')$ , and construct an  $F$  for which the equation has a solution which is not a minimiser.
2. (Iranian competition) Let  $f$  be continuous on  $[a, b]$  with  $f(a) = f(b) = 0$ , twice differentiable inside with  $M = \sup |f''| < \infty$ . Prove that  $\int_a^b |f(x)| dx \leq \frac{M(b-a)^3}{12}$  and describe the extremal function.
3. Solve Dido’s problem: among the curves of a given length joining two points of a line, find the one enclosing the largest area, using the multiplier of the isoperimetric constraint.
4. Minimise  $\int_0^1 ((y')^2 + y^2) dx$  with  $y(0) = 0$  and  $y(1) = 1$ , and then solve the same problem with the additional constraint  $\int_0^1 y = 0$ . Explain how the multiplier changes the shape of the answer.
5. Prove Legendre’s necessary condition  $F_{y'y'} \geq 0$  along a minimiser, and give a functional for which the Euler–Lagrange equation has a solution while the infimum equals  $-\infty$ .
6. Prove that the geodesics of the plane are straight lines and those of the sphere are great circles, in both cases by the Euler–Lagrange equation in a suitable parametrisation.

7. (Proposed problems) A parabola  $P$  has focus  $F$  at distance  $d$  from the directrix. Find the maximum length of an arc of  $P$  corresponding to a chord of length  $L$ .
8. Show that  $\inf \int_0^1 (1 - (y')^2)^2 dx$  over  $y$  with  $y(0) = y(1) = 0$  equals zero and is not attained, and say precisely which hypothesis of the direct method fails.
9. Among all  $C^1$  functions with  $f(0) = f(1) = 0$  and  $\int_0^1 (f')^2 = 1$  maximise  $\int_0^1 f$ , and identify the answer with the first eigenfunction of the Dirichlet problem on  $[0, 1]$ .
10. Prove the transversality condition for a problem with one free endpoint lying on a given curve, and use it to find the shortest path from a point to a smooth curve, recovering the orthogonality of the optimal segment.